

Body Fluids and Circulation

1. An artificial pacemaker is required in a person, when

- (1) AV valves are partially damaged
- (2) Heart beat is 70-72 times/min at the resting state
- (3) Atrioventricular bundles are conducting impulses properly
- (4) SA node gets damaged and activity of AV node is also suppressed

2. Given below are two statements: One is labelled as Assertion → 1st and 2nd heart sounds are systolic and diastolic sounds respectively

Reason → 1st and 2nd heart sounds are produced due to closure of valves

In the light of the above statements, choose the most appropriate answer from option given below

- (1) Both Assertion and Reason are correct and Reason is correct explanation of assertion
- (2) Both assertion and reason are correct but reason is not the correct explanation of assertion
- (3) Assertion is correct but reason is incorrect
- (4) Both assertion and reason are incorrect

3. Which one of the following is correct?

- (1) Lymph = Plasma + RBCs + WBCs
- (2) Blood = Plasma + RBCs + WBCs Platelets
- (3) Plasma = Blood – Lymphocytes
- (4) Serum = Blood + Fibrinogen

4. Which of the following is often referred to as atherosclerosis?

- (1) Angina
- (2) High blood pressure
- (3) Heart failure
- (4) Coronary artery disease

5. Read the following statements w.r.t. humans.

Statement A: ECG does not record heart sounds and blood pressure.

Statement B: Electrocardiogram is a graphical representation of the electrical activity of the heart.

Choose the correct option.

- (1) Both statements A and B are correct
- (2) Both statements A and B are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

6. If a person has systolic BP = 130 and pulse pressure = 80, then it is the case of :

- (1) Hypertension
- (2) Hypotension
- (3) Increased heart beat
- (4) High GFR

7. Match List-I with List-II to find out the correct option.

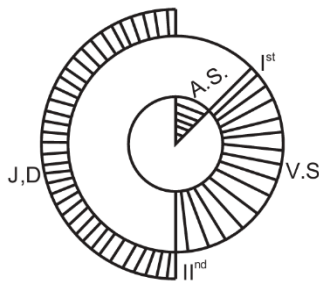
List-I	List-II
I. Fibrinogen	A. Osmotic balance
II. Globulin	B. Blood clotting
III. Albumin	C. Defense mechanism
IV. Proteins	D. 6-8% of total plasma

- (1) I-A, II-C, III-B, IV-D
- (2) I-A, II-B, III-C, IV-D
- (3) I-C, II-B, III-A, IV-D
- (4) I-B, II-C, III-A, IV-D

8. Angina pectoris is a symptom of :

- (1) Heart failure
- (2) Heart attack
- (3) Cardiac arrest
- (4) Arrhythmia

9. For how much time the AV valve remains opened.



- (1) 0.3 sec.
- (2) 0.7 sec.
- (3) 0.5 sec.
- (4) 0.8 sec.

10. In a healthy human, the rhythmic contractile

activity of the heart is initiated and maintained by the nodal tissue, present in/throughout the

- (1) Atrio-ventricular septum
- (2) Right upper corner of the right atrium
- (3) Lower left corner of the right atrium
- (4) Ventricular musculature

11. A person with B-ve blood group is hospitalized due to severe blood loss. For the treatment, he should be transfused with _____ blood

group due to the _____ on the donor's RBCs.

Select the correct option to fill in the respective blanks.

- (1) AB+ve , presence of A and B antigens
- (2) O-ve , absence of A and Rh antigens
- (3) A+ve , presence of A and absence of B antigens
- (4) O+ve , presence of Rh antigens

12. For safe blood transfusion;

- (1) donor's RBCs should not contain any antibodies against recipient's serum.
- (2) recipient's serum should not contain any antigens against donor's antibodies.
- (3) recipient's serum should not contain any antibodies against the antigens present on the surface of the RBCs of donors.
- (4) recipient's RBCs should not contain any antibodies against donor's antigens.

13. Match the column-I with column-II

	Column-I		Column-II
I.	Pericardium	A.	Upper chambers
II.	Atria	B.	Guard the opening between left atrium and left ventricle
III.	Tricuspid valve	C.	Protective membranes of heart
IV.	Bicuspid valve	D.	Guard the opening between right atrium and right ventricle

Choose the most appropriate answer from option given below

- (1) I-A, II-B, III-D, IV-C
- (2) I-C, II-D, III-A, IV-B
- (3) I-B, II-D, III-C, IV-A
- (4) I-C, II-A, III-D, IV-B

14. Select the granulocytes which constitute the least percentage of WBCs in a healthy individual.

- (1) Eosinophils
- (2) Monocytes
- (3) Neutrophils
- (4) Basophils

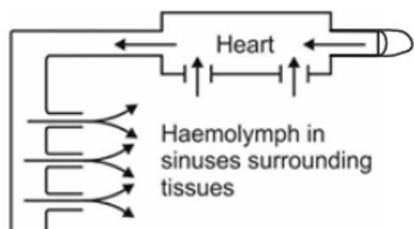
15. By about what percentage, the flow of blood increases from atria into the ventricles during atrial systole, per cardiac cycle in humans?

- (1) 70%
- (2) 30%
- (3) 50%
- (4) 90%

16. In the circulatory system of fishes

- (1) Atrium receives oxygenated blood only
- (2) Ventricle pumps only oxygenated blood
- (3) Deoxygenated blood passes through atrium and ventricle
- (4) Ventricle receives only oxygenated blood

17.



The schematic diagram given above best represents the circulatory system of

- (1) Arthropods like cockroach
- (2) Annelids like earthworms
- (3) Primates like human beings
- (4) Chordates like fish

18. Given below are two statements:

Assertion (A) : Blood coagulation is a cascade process.

Reason (R) : Coagulation is linked by a series of linked enzymatic reactions involving a number of factors in plasma.

In the light of the above statements, choose the most appropriate answer from option given below

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, and Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

19. Which of the following is a false statements ?

- (1) ECG is of a great clinical significance
- (2) Electrocardiograph is the recording of electrical changes during the cardiac cycle
- (3) To obtain a standard ECG, a patient is connected to the machine with 3 electrical electrodes (one to each wrist and to the left ankle)
- (4) Normal activities of the heart are regulated intrinsically

20. How many of the structures mentioned in the box below is/are directly in contact with deoxygenated blood in humans?

Mitral valve, pulmonary semilunar valve, aortic-semilunar valve, tricuspid valve, pulmonary artery

Choose the correct option.

- (1) Five
- (2) Four
- (3) One
- (4) Three

21. Which of the following statement is correct :

- (1) Blood groups are defined on the basis of antibodies present on Donan's membrane.
- (2) A person having 'A' blood group has anti –A antibody in plasma
- (3) A person having O+ blood group can donate blood to a person having 'A–' blood group.
- (4) 'O–' is the universal donor because of absence of Rh antigen on Donan's membrane

22. Which of the following events of human cardiac cycle occurs during the joint diastole?

- (1) Closure of mitral valves
- (2) Production of 'dub' heart sound
- (3) Complete ventricular filling
- (4) Production of the first heart sound

23. Given below are two statements:

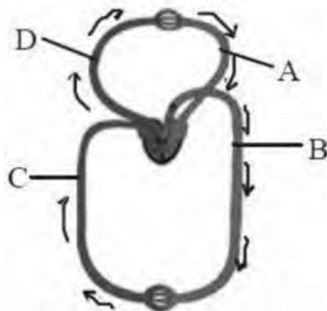
Assertion (A): In Struthio, closed circulatory system is present.

Reason (R): In Struthio, blood is carried in a closed network of arteries, veins, arterioles, venules and capillaries.

Choose the correct option.

- (1) Both (A) and (R) are true and (R) correctly explains (A)
- (2) Both (A) and (R) are false
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are true but (R) does not explain (A) correctly

24. The given figure shows a schematic plan of blood circulation in humans with labels A to D. Identify the label along with their functions and select the correct option.



- (1) C-Vena cava takes blood from body parts to right auricle
 (2) D-Dorsal aorta takes blood from heart to body parts
 (3) A-Pulmonary vein takes impure blood from body parts
 (4) B-Pulmonary artery takes blood from heart to lungs

25. Choose the correct set of blood cells which are involved in phagocytosis.

- (1) T-lymphocytes and B-lymphocytes
 (2) Neutrophils and Monocytes
 (3) Monocytes and Eosinophils
 (4) Thrombocytes and Neutrophils

26. Which of the following events do not occur during joint diastole ?

- I. All 4 chambers of heart are in relaxed state
 II. Tricuspid and bicuspid valves open
 III. Action potential is conducted from SAN to AVN
 IV. Blood from the pulmonary veins and vena cava flows into the left and right ventricles respectively through the left and right atria
 V. The Semilunar valves are closed

- (1) Only V
 (2) Only III
 (3) Only IV
 (4) Only I and II

27. Identify whether the given statements are true or false for double circulation.

I. It checks the mixing of oxygenated and deoxygenated blood.

II. It carries only oxygenated blood.

Choose the correct option accordingly.

- (1) I - False, II- False
 (2) I-True, II-True
 (3) I - False, II- True
 (4) I-True, II-False

28. The world famous ultrafit athlete known for his running record underwent a medical examination and it was observed that his heart rate at resting stage was about 65 beats/min. The most reasonable explanation for this is, that

- (1) He suffers from a defect in the pacemaker of his heart
 (2) He has higher stroke volume
 (3) He has lower stroke volume
 (4) There is an anomaly in the reading as athletes can never have such low heart rate

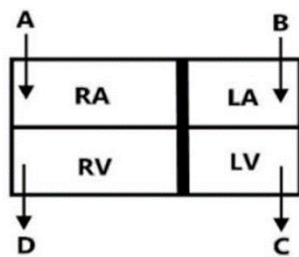
29. Pulmonary vein and pulmonary artery in humans differ from each other in all of the following, except

- (1) Thickness of their selective tunica media
 (2) Partial pressure of oxygen in the blood that they carry
 (3) Number of layers in their walls
 (4) Diameter of their lumen

30. A patient went to see a cardiologist complaining about being tachycardic. Upon examination, his heart rate was found to be 90 beats per minute; his end-diastolic volume was found to be 140 mL whereas his end-systolic volume was 60 mL w.r.t. left ventricle. What could be his cardiac output per min?

- (1) 4.4 litres
 (2) 5.8 litres
 (3) 6.6 litres
 (4) 7.2 litres

31. Identify A, B, C and D in the given table.



- (1) A - Pulmonary vein, B - Vena cava, C - Pulmonary artery, D - Aorta
- (2) A - Pulmonary artery, B - Pulmonary vein, C - Vena cava, D - Aorta
- (3) A - Aorta, B - Pulmonary artery, C - Pulmonary vein, D - Vena cava
- (4) A - Vena cava, B - Pulmonary vein, C - Aorta, D - Pulmonary artery

32. Given below are two statements:

Statement-I: Systemic heart refers to left atrium and left ventricle.

Statement-II: Systemic circulation provides nutrients, O₂ and other essential substances to the tissues.

Choose the correct option

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

33. Cardiac output is;

- (1) volume of the blood pumped out by each ventricle per minute.
- (2) volume of the blood contained in the entire heart.
- (3) volume of the oxygenated blood pumped by heart.
- (4) volume of the deoxygenated blood pumped by heart.

34. Read the following statements w.r.t. circulation of blood in human body and choose the correct option.

Statement A: Lymph is an important carrier for nutrients and hormones.

Statement B: Tissue fluid has the same distribution of all substances as present in blood plasma.

In the light of the above statements, choose the most appropriate answer from option given below

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

35. Every 300 mL of oxygenated blood can deliver around ____ of O₂ to the tissues, under normal physiological conditions in humans.

Select a suitable option to fill the blank.

- (1) 5 mL
- (2) 15 mL
- (3) 25 mL
- (4) 35 mL

36. Which of the following statement is incorrect?

- (1) Heart is endodermal in origin.
- (2) Human heart is situated in between the two lungs.
- (3) Heart is a double walled membranous bag.
- (4) Human heart has two atria and two ventricles.

37. Arteries are best defined as the vessels which;

- (1) supply oxygenated blood to the different organs.
- (2) carry blood away from the heart to different organs.
- (3) break up into capillaries which reunite to form a vein.
- (4) carry blood from one visceral organ to another visceral organ.

38. Read the following statements.

- I. Systemic aorta originates from left ventricle and distributes deoxygenated blood to lungs.**
- II. Pulmonary arch originates from right ventricle and carries oxygenated blood to various body parts except lungs.**
- III. Coronary veins return deoxygenated blood from heart wall to right auricle.**

Which of the following statement(s) correctly identifies the function of blood vessels associated with human heart?

- (1) Only III
- (2) I and II
- (3) I, II and III
- (4) I and III

39. If a person goes to higher altitudes and stays there for few days, then what change is usually seen in his formed elements?

- (1) Decrease in number of RBCs
- (2) Increase in thrombocyte count
- (3) Increase in size of RBCs
- (4) Increase in number of RBCs

40. In humans, the role of pleural fluid present in between the pleural membranes is to

- (1) Increase the friction on the lung surface
- (2) Stop the exchange of respiratory gases
- (3) Facilitate abnormal inflation of lungs
- (4) Reduce the friction on the lung surface

41. A special arrangement of conducting system in human heart is such that AV node delays the passage of cardiac impulses for about 0.1 sec from atria to ventricles. This permits/allows

- (1) Atria to empty blood to aorta directly
- (2) Complete emptying of ventricles
- (3) Atria to get empty completely
- (4) Partial contraction of atria

42. A red blood cell, entering the right side of the heart passes by or through the following structures

- i. Atrioventricular valves**
- ii. Semi-lunar valves**
- iii. Right atrium**
- iv. Right ventricle**
- v. SAN**

Which of the following sequence is correct ?

- (1) ii → iii → i → iv → v
- (2) v → i → v → ii → iv
- (3) iii → v → i → ii → iv
- (4) v → iii → i → iv → ii

43. If due to some injury the chordae tendineae of the tricuspid valve of the human heart is partially non-functional, what will be the immediate effect?

- (1) The flow of blood into the aorta will be slowed down.
- (2) The pacemaker will stop working.
- (3) The blood will tend to flow back into the left atrium.
- (4) The flow of blood into the pulmonary artery will be reduced.

44. Which of the following correctly explains a phase/event in cardiac cycle in a standard electrocardiogram?

- (1) QRS complex indicates atrial contraction.
- (2) QRS complex indicates ventricular contraction.
- (3) Time between S and T represents atrial systole.
- (4) P-wave indicates beginning of ventricular contraction.

45. The deposition of lipids on the wall lining the lumen of large and medium sized arteries is referred to as;

- (1) Osteoporosis
- (2) Osteoarthritis
- (3) Atherosclerosis
- (4) Deep vein thrombosis

46. Select the correct option represents the events occurs between 2nd and 1st heart sound :

- A.→ Joint diastole
- B.→ Auricular systole
- C.→ Blood fills into both auricles and ventricles
- D.→ Ventricular systole
- E.→ Blood from auricles fills into ventricles passively
- F.→ Blood from auricles fills into ventricles actively
- G.→ Time duration is 0.5 sec.
- H.→ AV valve remains closed

- (1) A, C, B, G, F, and E
- (2) A, B, C, G, G, and E
- (3) A, B, C, E, F, and G
- (4) A, F, C, G, B, and F

47. Which of the following represents the correct path of blood flow in humans?

- (1) Right ventricle to aorta
- (2) Pulmonary veins to left atrium
- (3) Inferior vena cava to left atrium
- (4) Left ventricle to pulmonary artery

48. Given below are two statements:

Assertion (A) : Heart attack is the condition when the heart muscles are suddenly damaged by an inadequate blood supply.

Reason (R) : Heart failure is the state of heart when it is not pumping blood effectively enough to meet the needs of the body.

Choose the correct option

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, and Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

49. Given below are two statements:

Statement-I: Electrocardiogram is a record or electrical activity of the heart which shows certain waves called P, Q, R, S and T waves.

Statement-II: Electrocardiogram gives important information concerning the spread of excitation signal to the different parts of heart and it is of value in the diagnosis of cases of abnormal cardiac rhythm and myocardial damage.

Choose the correct option

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but Statement II is incorrect.
- (3) Statement is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

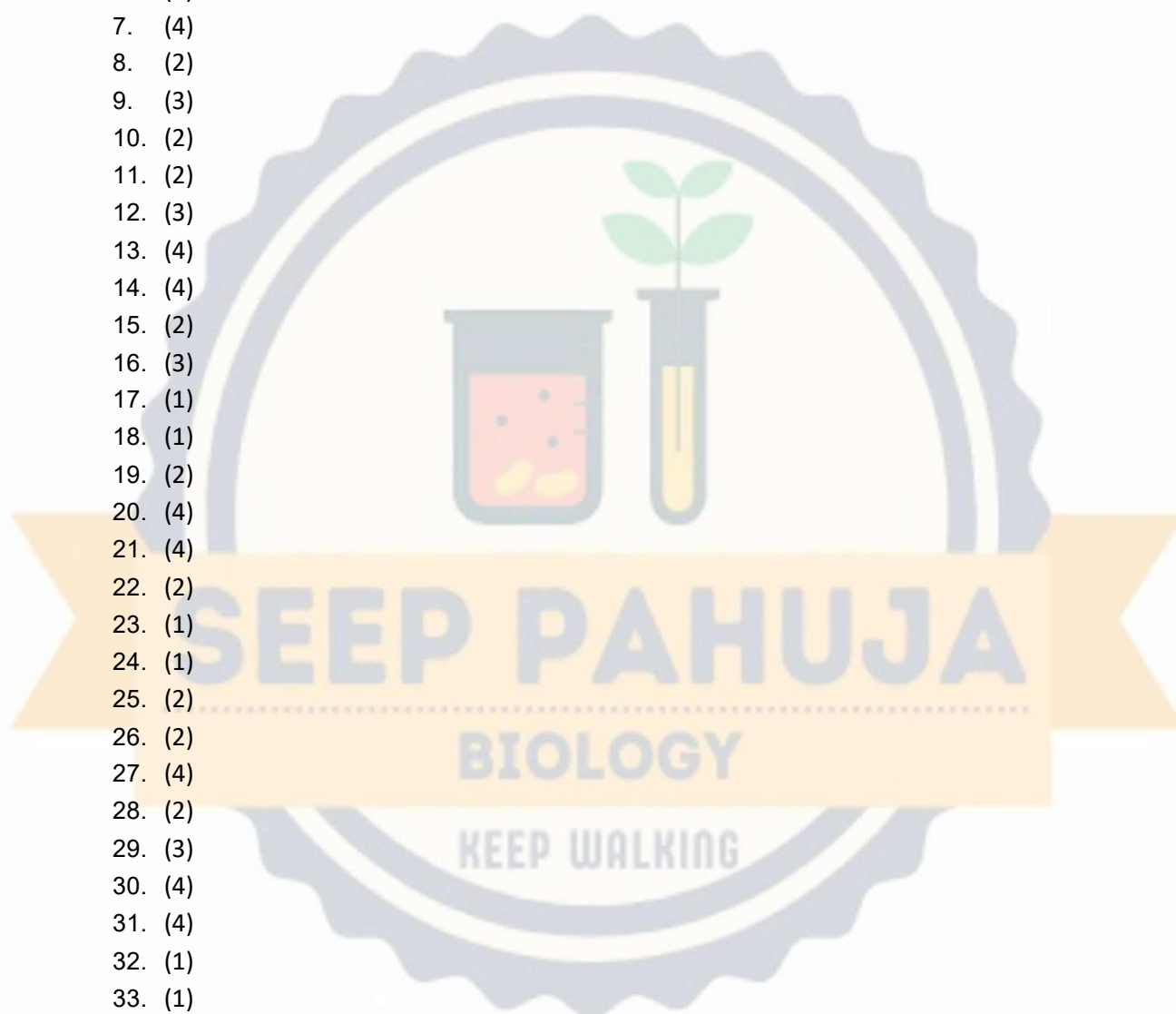
50. Which of the following statements are incorrect about lymph :

- (i) Lymph is colourful as it has haemoglobin but no RBC.
- (ii) Lymph has unidirectional flow.
- (iii) The fluid present in lymphatic system is called lymph.
- (iv) Lymph is important carrier for nutrients and hormones
- (v) Lymph contains specialized lymphocytes
- (vi) lymph has similar mineral distribution as that in plasma
- (vii) Lymph has absorbed digested fats

- (1) (i), (ii), and (vi)
- (2) (iii), (iv) and (vi)
- (3) (ii), (vi) and (vii)
- (4) only (i)

Answer Key

1. (4)
2. (2)
3. (2)
4. (4)
5. (1)
6. (2)
7. (4)
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42. (4)
43. (4)
44. (2)



- 45. (3)
- 46. (3)
- 47. (2)
- 48. (2)
- 49. (1)
- 50. (4)

